

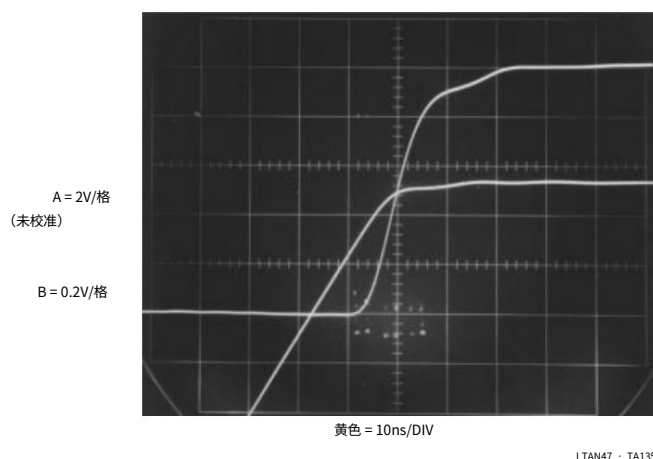


图 135. 时间高度转换器的电流源关断细节

要校准该电路,请输入 250ns 宽度的脉冲并调整 1k Ω 电位器以获得 10V 输出。该电路会将 20ns 至 250ns 的脉冲宽度转换为 1% 的典型精度。20ns 的最小可测量宽度是由于无法将 100pF 电容器完全放电。如果这令人反感,可以用饱和度较低的器件替换 Q4,或者可以抵消 A1 的输出。

真有效值宽带电压表

大多数交流 RMS 测量使用对数技术来计算波形的 RMS 值。该方法将带宽限制在 1MHz 以下,并将波峰因数性能限制在 10:1 左右。实际上,波形的 RMS 值定义为负载中的发热值。专用仪器采用基于热的组件来计算输入的 RMS 值。与基于对数的转换器相比,热方法提供了显着改进的带宽和波峰因数能力。

热 RMS-DC 转换器是直接作用热电子模拟计算机。热技术是明确的,依赖于第一原理。简单的操作允许使用基于对数计算的隐式间接方法无法实现的宽带性能。

图 136 显示了实现基于热的 RMS-DC 转换器的经典方案。这里,直流放大器迫使第二个相同的加热器传感器对达到与输入驱动对相同的热条件。这种差分传感、反馈强制环路使环境温度变化成为共模项,从而消除了它们的影响。此外,尽管电压和热

相互作用是非线性的,输入输出电压关系与单位增益是线性的。这种布置抑制环境温度变化的能力取决于加热器-传感器对是否等温。这可以通过以远低于环境变化的时间常数对它们进行隔热来实现。如果加热器-传感器对的时间常数匹配,则环境温度项将在相位和幅度上同等地影响这些对。

直流放大器将抑制该共模项。请注意,虽然这些对是等温的,但它们彼此绝缘。线对之间的任何热相互作用都会降低系统的热增益项。

这会导致不利的信噪比性能,限制动态工作范围。图 136 的输出是线性的,因为匹配热对的非线性电压-温度关系相互抵消。

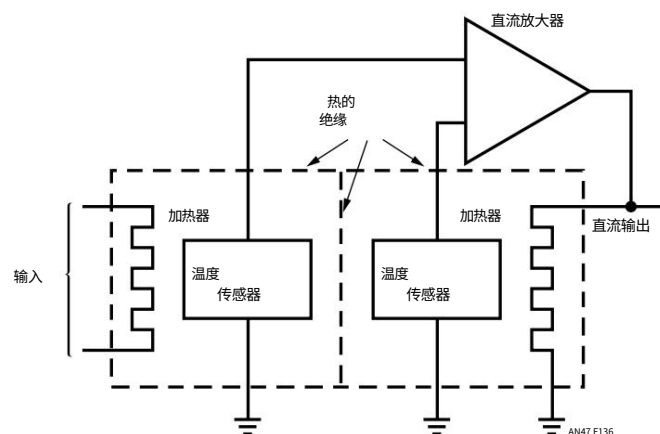


图 136. 概念热 RMS-DC 转换器

这种方法的优点使其在基于热的 RMS-DC 测量中广受欢迎。通常,该组件由匹配的加热电阻器、传感器和隔热材料组成。这些组件相对较大并且生产成本昂贵。图 137 的经济型宽带热电压表基于单片热转换器。LT1223 提供增益,并驱动 LT1088 RMS-DC 热转换器。19 LT1088 的温度检测二极管由电源偏置。A1 设置为增益为 9000 的差分伺服放大器,提取二极管的差分信号并对 Q1 进行偏置。Q1 驱动 LT1088 的一个加热器,完成一个循环。3300pF 电容器提供稳定的滚降。

注 19:有关该器件的完整详细信息以及有关热转换注意事项的讨论请参见参考文献 40。

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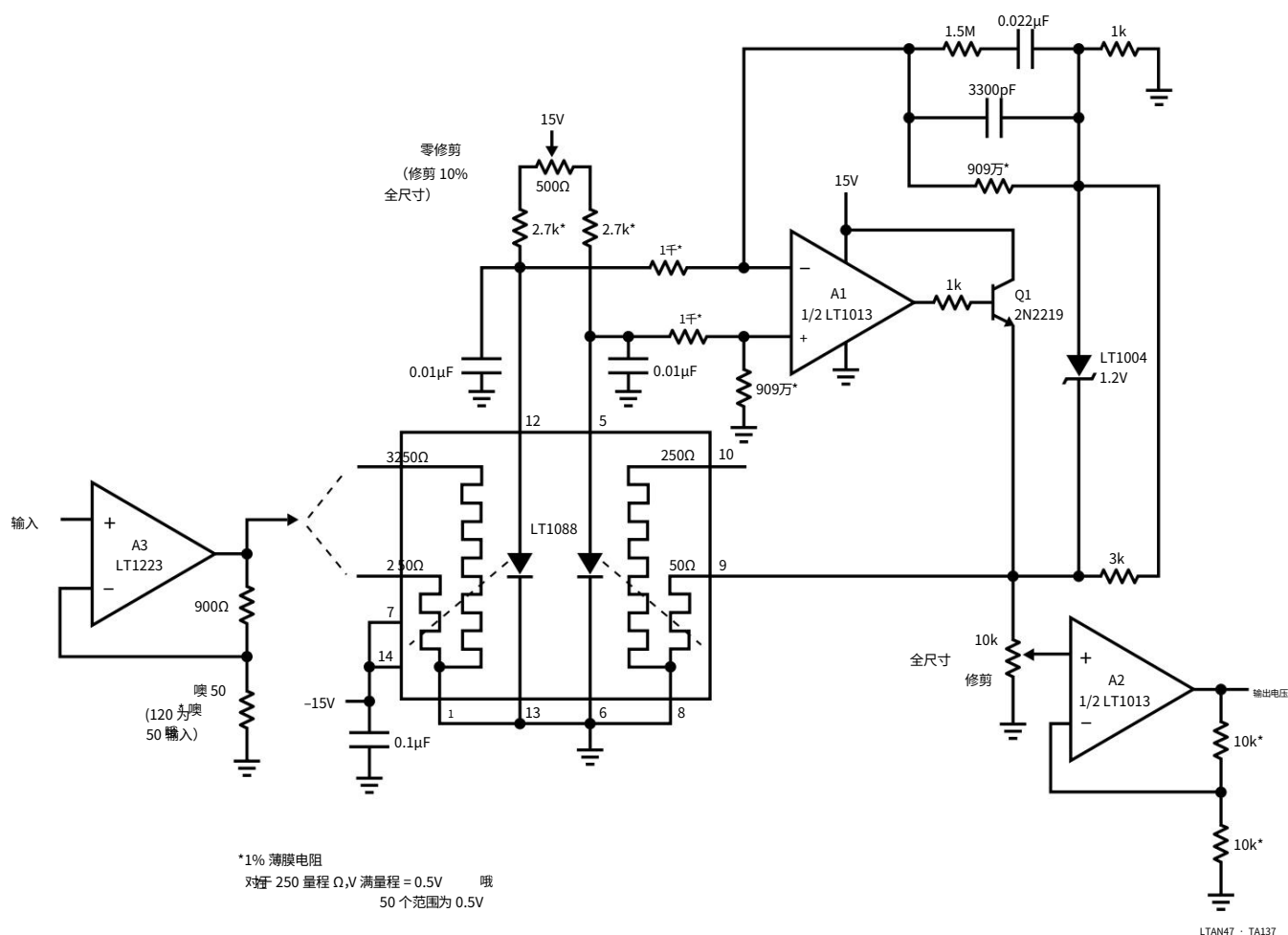


图 137. 宽带真有效值电压表

1.5M-0.022μF 组合通过降低输出转换期间的增益来改善稳定。LT1088 的平方律增益意味着小输入时的整体环路增益较低。通常,这会导致数值低于刻度的 10%-20% 时缓慢稳定。LT1004 1k-3k 网络是一个简单的断点,可提高该区域的放大器增益以改善稳定性。A2 是增益可调输出级,用于补偿 LT1088 两侧的增益变化。要调整电路,请输入大约 10% 刻度的直流信号(例如 0.05V)。调整调零,使 $V_{OUT} = V_{IN}$ 。接下来,应用满量程直流输入并将满量程微调设置为输出处的该值。

重复调整,直到两者都固定在满量程的 1% 以内。另一种调整方案包括不施加输入,将 Q1 的基极接地并设置零调整,直到 A1 的输出有效。然后,将 Q1 的基极接地,施加满量程输入,并在输出端针对该值进行满量程调整。

图 138 是误差与输入频率的关系图。LT1088 的额定工作频率为 2% 至 100MHz (50Ω 加热器)或 1% 至 20MHz (250Ω 加热器)。因此,显示的大部分错误是由于 A3 中的带宽限制造成的,但性能仍然令人印象深刻。这些图包括在两个加热器的不同输入水平下获取的数据。250Ω 的 500mV 输入在 8MHz 时下降至 1%,在 14MHz 时下降至 2.5%,然后峰值严重超过 17MHz。该输入电平迫使 A3 输出 9.5 VRMS,从而引入大信号带宽限制。250Ω 加热器的 400mV 输入在 20MHz (LT1088 的 250Ω 加热器规格限制)下显示出基本平坦的结果。

尽管 A3 的 50mA 输出将最大输入限制为约 100mVRMS (LT1088 为 1.76VRMS),50Ω 加热器可提供明显更宽的带宽。

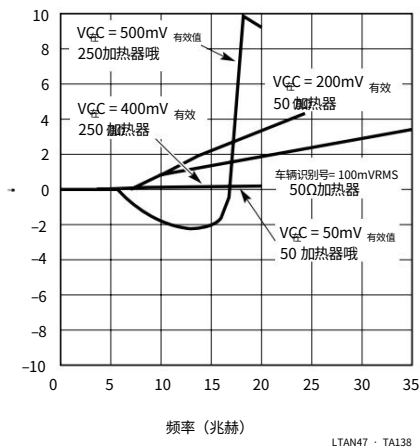


图 138. RMS 电压表的精度图

应用程序第四部分 - 其他
电路

射频调平环

图137的宽带交流变换可以应用于其他领域。常见的射频要求是根据输入、时间和温度的变化稳定波形的幅度。仪器和发射机经常需要此功能,如果必须保持波形纯度,这并不容易。图 139A 显示了 25MHz RF 电平环路。RF 输入应用于 AD539 宽带乘法器。乘法器的输出驱动A1。

A1 的输出由基于 LT1088 的 RMS DC 转换器转换为 DC (参见前面的电路)。伺服放大器将此输出与可设置的直流基准进行比较,并对乘法器的控制通道进行偏置,从而完成一个循环。
0.33μF 电容器通过在远低于 LT1088 伺服系统响应的频率下滚降增益来提供频率补偿。该环路将输出的 25MHz RMS 幅度保持在 DC 参考值。负载、输入、电源和其他变量的变化被拒绝。

图 139B 是一个类似的电路,尽管性能不太好,但成本显着降低。RF 输入应用于运算跨导放大器 LT1228 A1。A1 的输出为电流反馈放大器 LT1228 A2 供电。A2 的输出 (即电路的输出)由基于 A3 的增益控制配置进行采样。

这种布置类似于图112和图114中描述的增益控制环路,在A1处闭合增益控制环路。4pF电容补偿整流二极管

电容,增强输出平坦度与频率的关系。A1 的 ISET 输入电流控制其增益,从而实现整体输出电平控制。尽管输出漂移、失真和调节比以前的电路稍高,但这种射频调平方法既简单又便宜。

压控电流源

图 140 显示了带有负载和参考地的控制电压的压控电流源。这个简单而强大的电路根据控制电压的符号和幅度产生输出电流。该电路的比例因子由电阻器 R 设置。A1 由 VIN 偏置,驱动电流流过 R (在本例中为 10Ω)和负载。

A2 通过 R 进行差分感应,闭合返回 A1 的环路。负载电流是恒定的,因为 A1 的环路强制 R 上有固定电压。2k-100pF 组合设置滚降,配置稳定。图 141 显示动态响应。迹线 A 是电压控制输入,迹线 B 是输出电流。响应快速、干净,延迟为 5 纳秒,无转换残留或畸变。

高功率压控电流源

图 142 与基本电流源相同,只是它添加了 1A 升压级 (改编自图 104)以增加输出功率。在 A1 的反馈环路中加入升压器可消除其直流误差。请注意,由于该电路固有的工作电流限制性质,升压器的电流限制功能已被删除。图 143 显示该电路的响应与低功耗版本一样干净,但延迟慢了约 20ns。值得一提的是,将 A2 升压器置于 A1 的反馈路径中涉及的环路稳定性考虑非常重要。该电路在附录 C “振荡问题 - 无泪频率补偿”中进行了处理。

和

18ns 断路器

图 144 显示了一个简单电路,该电路将在负载中的电流超过预设值 18ns 后关闭。该电路已用于在开发探测期间保护集成电路,并且还可用于在微调 and 校准期间保护昂贵的负载。由于负载的一侧接地,因此电路的多功能性得到增强。正常情况下,Q1 的发射极 (迹线 A,图 145,是 Q1 的电流,迹线 C 是其电流)

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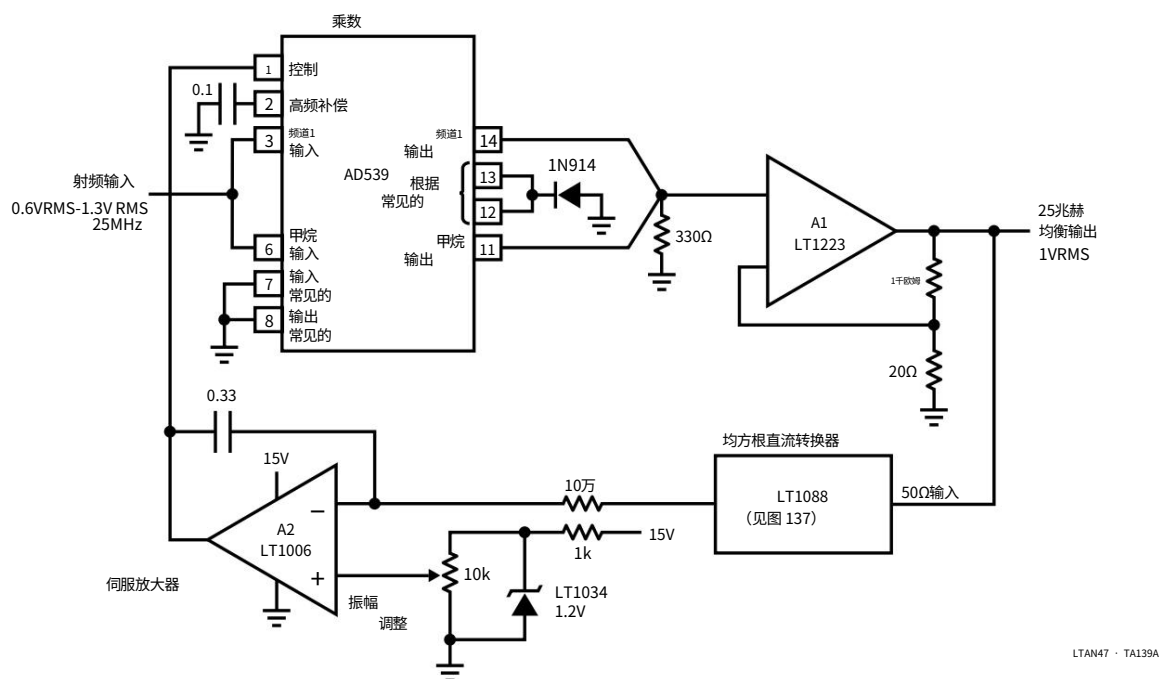


图139A.射频调平环

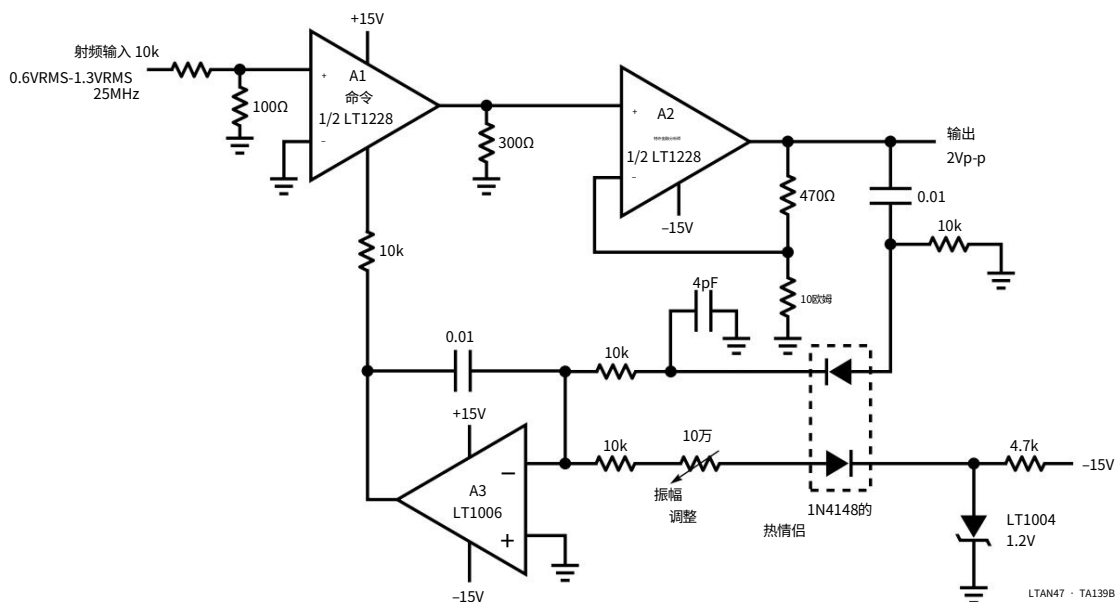


图139B.简单的射频调平环路

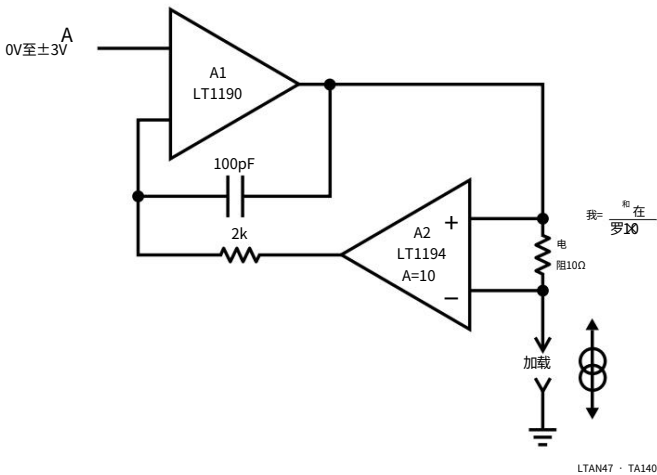


图 140. 具有接地负载的快速、精确、压控电流源

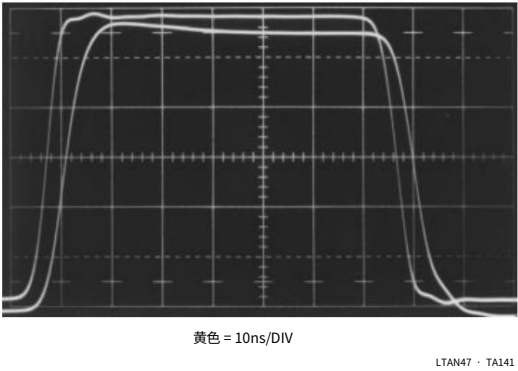


图 141. 电流源的动态响应。延迟为 4 纳秒, 稳定稳定

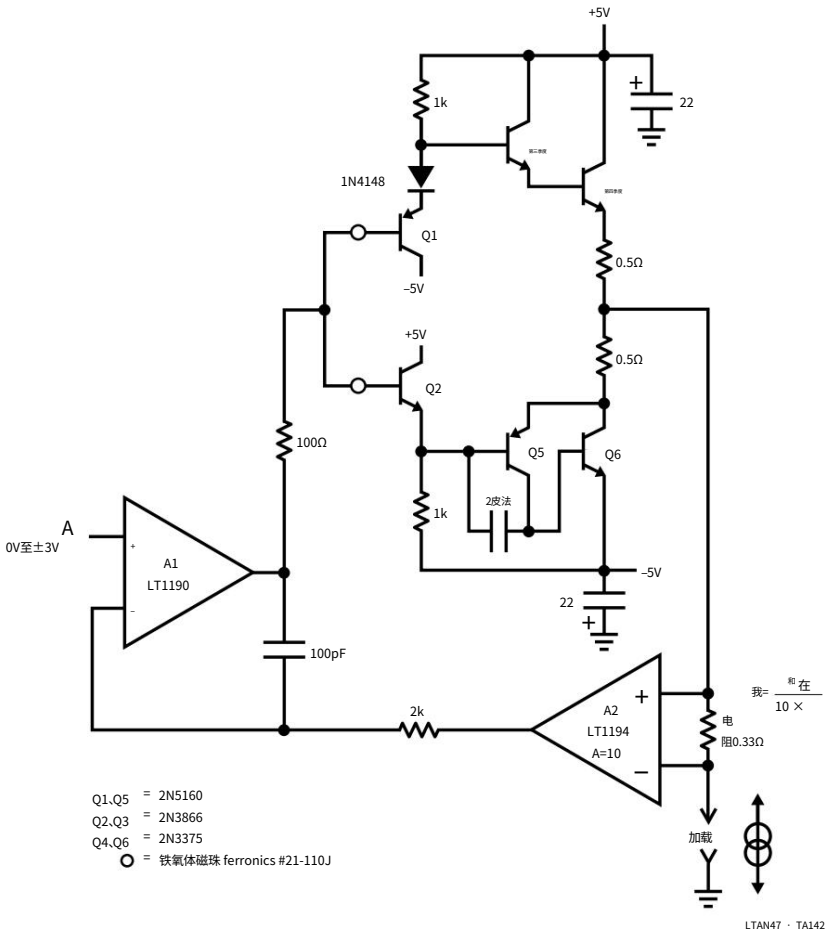


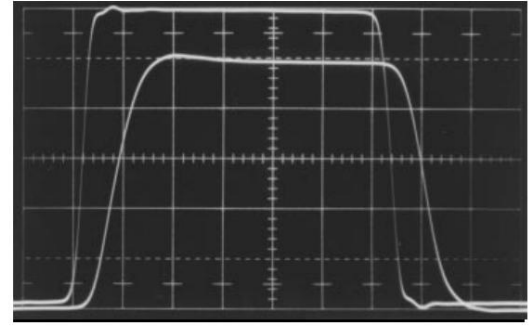
图 142. 高功率、宽带压控电流源

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电压)被偏置,通过 10Ω 分流器向负载供电。差分放大器 A1 的输出低于比较器 A2 的电压编程跳变点,并且 Q2 关闭。当发生过载时,Q1 的发射极电流开始增加(迹线 A,就在第三个垂直分区之前)。A1 的输出(迹线 B)随着 10Ω 分流器电压的增加而开始上升。 $9k-1k$ 分压器将 A1 输入保持在其共模范围内。

同时,Q1 的发射极电压(迹线 C)开始下降,达到 beta 限制。当 A1 的负载电流超过 A2 的跳变点时,A2 (迹线 D)变高,从而导通 Q2。Q2 的开启窃取了 Q1 的基极驱动,从而关闭负载

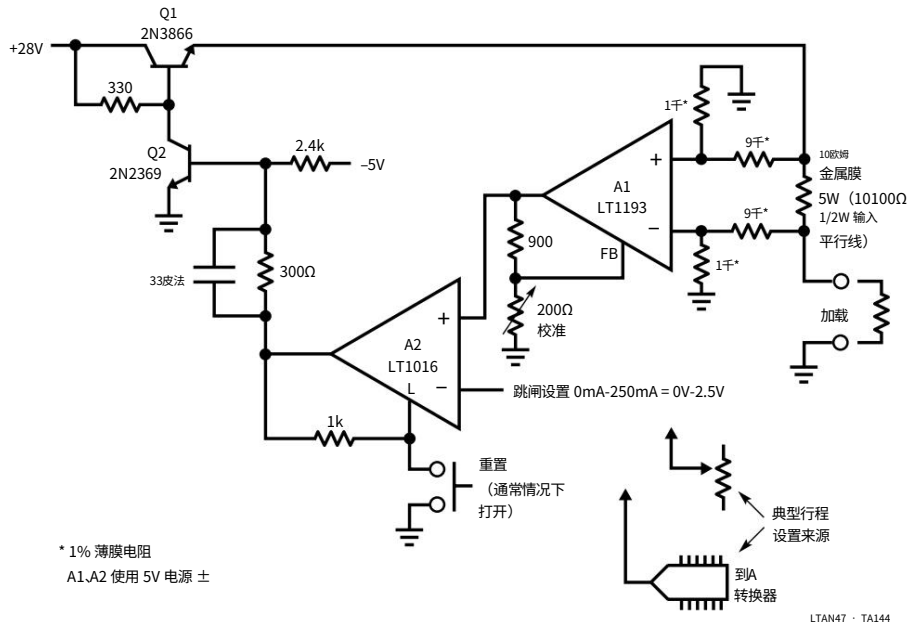
A = 0.5V/格
B = 0.2A/格



黄色 = 20ns/DIV

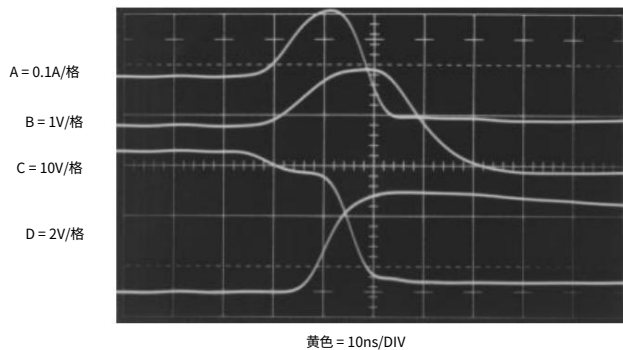
LTAN47 · TA143

图 143. 高功率电流源的 1A 脉冲响应



LTAN47 · TA144

图 144. 具有电压可编程跳变点的 18ns 断路器



LTAN47 · TA145

图 145. 18ns 断路器的工作波形。
输出电流(迹线 A)开始上升后 18ns 电路输出(迹线 C)关闭

当前的。A2 锁存引脚处的本地正反馈导致其锁存于该关闭状态。当负载故障被清除后,按钮可用于重置电路。

从出现过大负载电流到完全关闭的延迟在 18ns 以内。解释波形时应考虑迹线 A 电流探头的 4ns 延迟。要校准该电路,请将 Q2 的基极接地并安装 250mA 负载。调整 200Ω 微调以在 A1 处提供 2.5V 输出。接下来,移除负载,将 Q2 的底座接地并按下重置按钮。最后,输入所需的跳闸设定电压,电路即可使用。

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笔记:本应用说明源自最初准备在应用说明上手稿

APPENDIX A

ABC's of Probes – Tektronix, Inc.

This appendix, guest written by the engineering staff of Tektronix, Inc., is a distillation of their booklet, "ABC's of Probes". The complete booklet is available, at no charge,

through any Tektronix sales office or call 800-835-9433 ext. 170. For excellent technical background on probe theory see Reference 42.

PART I: UNDERSTANDING PROBES

The vital link in your measurement system

Probes connect the measurement test points in a DUT (device under test) to the inputs of an oscilloscope. Achieving optimized system performance depends on selecting the proper probe for your measurement needs.

Though you could connect a scope and DUT with just a wire, this simplest of connections would not let you realize the full capabilities of your scope. By the same token, a probe that is not right for your application can mean a significant loss in measurement results, plus costly delays and errors.

Why not use a piece of wire?

Good question: There are legitimate reasons for using a piece of wire or, more correctly, two pieces of wire; some low bandwidth scopes and special purpose plug-in amplifiers only provide binding post input terminals, so they offer a convenient means of attaching wires of various lengths.

DC levels associated with battery operated equipment could be measured. Low frequency (audio) signals from the same equipment could also be examined. Some high output transducers could also be monitored. However, this type of connection should be kept away from line-operated equipment for two basic reasons, safety and risk of equipment damage.

Safety: Attachment of hookup wires to line-operated equipment could impose a health hazard, either because the "hot" side of the line itself could be accessed, or because internally generated high voltages could be contacted. In both cases, the hookup wire offers virtually no operator protection, either at the equipment source or at the scope's binding posts.

Risk of Equipment Damage:

Two unidentified hookup wires, one signal lead and one ground, could cause havoc in line-operated equipment. If the "ground" wire is attached to **any** elevated signal in line-operated equipment, various degrees of damage will result simply because both the scope and the equipment are (or should be) on the same three-wire outlet system, and short-circuit continuity is completed through one common ground.

Performance Considerations:

In addition to the hazards just mentioned, there are two major performance limitations associated with using hookup wires to transfer the signal to the scope: circuit loading and susceptibility to external pickup.

Circuit Loading: This subject will be discussed in detail later, but circuit loading by the test equipment (scope-probe) is a combination of resistance and capacitance. Without the benefit of using an attenuator (10X) probe, the loading on the device under test (DUT) will be 1M ohm (the scope input resistance) and more than 15 picofarad (15pF), which is the typical scope input capacitance plus the stray capacitance of the hookup wire.

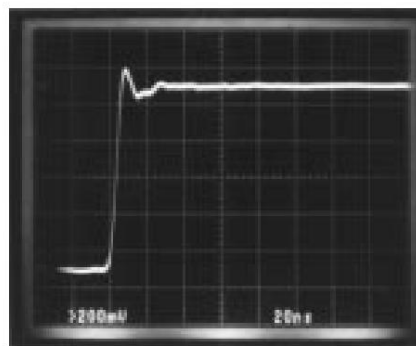


Figure 1-1

Figure 1-1 shows what a "real world" signal from a 500 ohm impedance source looks like when loaded by a 10M ohm, 10 pF probe:

the scope-probe system is 300MHz. Observed risetime is 6 nSec.

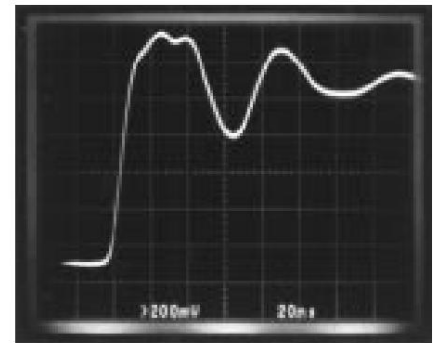


Figure 1-2

Figure 1-2 shows what happens to the same signal when it is accessed by two 2-meter lengths of hookup wire: loading is 1M ohm (the scope input resistance) and about 20 pF (the scope input capacitance, plus the stray capacitance of the wires). Observed risetime has slowed to 10 nSec and the transient response of the system has become unusable.

Susceptibility to External Pickup: An unshielded piece of wire acts as an antenna for the pickup of external fields, such as line frequency interference, electrical noise from fluorescent lamps, radio stations and signals from nearby equipment. These signals are not only injected into the scope along with the wanted signal, but can also be injected into the device under test (DUT) itself.

The source impedance of the DUT has a major effect on the level of interference signals developed in the wire. A very low source impedance would tend to shunt any induced voltages to ground, but high frequency signals could still appear at the scope input and mask the wanted signal. The answer, of course, is to use a probe which, in addition to its other features, provides coaxial shielding of the center conductor and virtual elimination of external field pickup.

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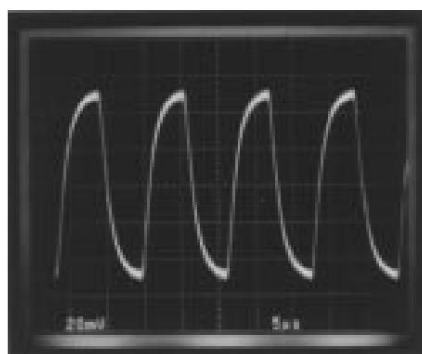


Figure 1-3

Figure 1-3 shows what a low level signal from a high impedance source (100mV from 100K ohm) looks like when accessed by a 300MHz scope-probe system. Loading is 10M ohm and 10 pF. This is a true representation of the signal, except that probe resistive loading has reduced the amplitude by about 1%: the observed high frequency noise is part of the signal at the high impedance test point and would normally be removed by using the BW (bandwidth) limit button on the scope. (See Figure 1-4.)

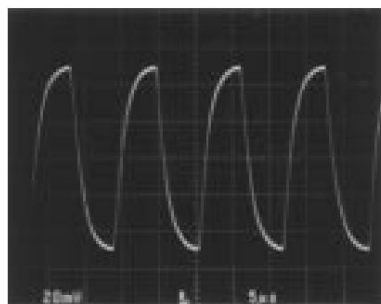


Figure 1-4

If we look at the same test point with our pieces of wire, two things happen. The amplitude drops due to the increased resistive and capacitive loading, and noise is added to the signal because the hookup wire is completely unshielded. (See Figure 1-5)

Most of the observed noise is line frequency interference from fluorescent lamps in the test area.

Probably the most annoying effect of using hookup wire to observe high frequency signals is its unpredictability. Any touching or rearrangement of the leads can produce different and nonrepeatable effects on the observed display.

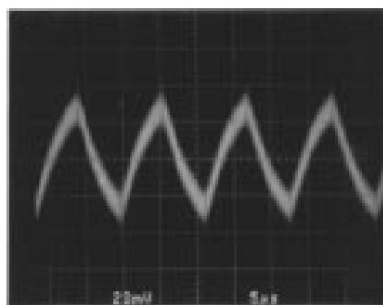


Figure 1-5

Benefits of using probes

Not all probes are alike and, for any specific application, there is no one ideal probe; but they share common features and functions that are often taken for granted.

Probes are convenient. They bring a scope's vertical amplifier to a circuit. Without a probe, you would either need to pick up a scope and attach it to a circuit, or pick up the circuit and attach it to the scope. Properly used, probes are convenient, flexible and safe extensions of a scope.

Probes provide a solid mechanical connection. A probe tip, whether it's a clip or a fine solid point, makes contact at just the place you want to examine.

Probes help minimize loading. To a certain extent, all probes load the DUT—the source of the signal you are measuring. Still, probes offer the best means of making the connections needed. A simple piece of wire, as we have just seen, would severely load the DUT; in fact, the DUT might stop functioning altogether.

Probes are designed to minimize loading. Passive, non-attenuating 1X probes offer the highest capacitive loading of any probe type—even these, however, are designed to keep loading as low as possible.

Probes protect a signal from external interference. A wire connection, as described earlier, in addition to loading the circuit, would act as an antenna and pick up stray signals such as 60Hz power, CBers, radio and TV stations. The scope would display these stray signals as well as the signal of interest from the DUT.

Probes extend a scope's signal amplitude-handling ability. Besides reducing capacitive and resistive loading, a standard passive 10X

(ten times attenuation) probe extends the on-screen viewability of signal amplitudes by a factor of ten.

A typical scope minimum sensitivity is 5V/division. Assuming an eight-division vertical graticule, a 1X probe (or a direct connection) would allow on-screen viewing of 40V p-p maximum. The standard 10X passive probe provides 400V p-p viewing. Following the same line, a 100X probe should allow 4kV on-screen viewing. However, most 100X probes are rated at 1.5kV to limit power dissipation in the probe itself.

Check the specs. Bandwidth is the probe specification most users look at first, but plenty of other features also help to determine which probe is right for your application. Circuit loading, signal aberrations, probe dynamic range, probe dimensions, environmental degradation and ground-path effects will all impact the probe selection process, as discussed in the pages that follow.

By giving due consideration to probe characteristics that your application requires, you will achieve successful measurements and derive full benefit from the instrument capabilities you have at hand.

How probes affect your measurements

Probes affect your measurements by loading the circuit you are examining. The loading effect is generally stated in terms of impedance at some specific frequency, and is made up of a combination of resistance and capacitance.

Source Impedance. Obviously, source impedance will have a large impact on the net effect of any specific probe loading. For example, a device under test with a near zero output impedance would not be affected in terms of amplitude or risetime to any significant degree by the use of a typical 10X passive probe. However, the same probe connected to a high impedance test point, such as the collector of a transistor, could affect the signal in terms of risetime and amplitude.

Capacitive Loading. To illustrate this effect, let's take a pulse generator with a very fast risetime. If the initial risetime was assumed to be zero ($t_r = 0$), the output t_r of the generator would be limited by the